

Multilevel Selection, Hierarchical Causation, and Constitutional Lock-in: Reconciling Gould with Gene-Centered Memetics

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March 2026 | Version 1.0

Abstract

This article integrates three apparently incompatible theoretical frameworks: Gould's (2002) multilevel evolutionary hierarchy, Okasha's (2024) distinction between organizational and selection hierarchies, and gene-centered memetic theory applied to law (Lerer, 2025). I argue that (1) type 2 multilevel selection (MLS2) lacks causal microfoundations when applied to legal systems; (2) higher-level emergent effects are real but operate as downward constraints, not as selection units; and (3) the Constitutional Lock-in Index 3.0 (CLI 3.0) mathematically formalizes these constraints through the Birkhoff polytope. Preliminary empirical validation with six Latin American countries (1983–2025) shows that CLI 3.0 discriminates successful from failed regimes with 78% accuracy. The analysis reconciles Dawkins's rejection of group selection with Gould's recognition of irreducible macroevolutionary processes, providing quantitative foundations for comparative institutional analysis.

Keywords: *multilevel selection, evolutionary hierarchy, legal memetics, extended phenotype, constitutional lock-in, lattice theory, Gould, Okasha, Dawkins*

JEL: K10, B52, D02, C73

1. Introduction: The Evolutionary Hierarchy Dilemma in Law

1.1 The Conceptual Problem

Applying evolutionary concepts to legal phenomena confronts an unresolved tension whose consequences extend well beyond theoretical aesthetics. Stephen Jay Gould (2002), in *The Structure of Evolutionary Theory*, his 1,400-page magnum opus, argued that natural selection operates at multiple hierarchical levels: genes, organisms, demes, species, and clades, each exhibiting emergent properties irreducible to lower levels. Gould treated species as “darwinian individuals” capable of differential reproduction through speciation and extinction, insisting that macroevolutionary patterns could not be explained by microevolutionary processes alone. Richard Dawkins (1976, 1982), by contrast, maintained that only replicators, genes in biology and memes in culture, qualify as genuine selection units. Higher levels are vehicles, interactors, or aggregate effects; they participate in the evolutionary process but are not its units.

For forty years this debate remained largely verbal, a dispute between two positions that appeared irreconcilable. The philosophical literature, particularly Sober and Wilson (1998), Okasha (2006), and Godfrey-Smith (2009), refined the conceptual territory without producing decisive empirical tests. This article argues that comparative law provides such a test, and that the results favor a specific synthesis.

The tension becomes an empirical question in the legal domain. When a constitutional system persists for decades while resisting reform, as Argentina’s Article 14bis has done for 68 years across 23 failed reform attempts, are we observing:

- (a) *Higher-level selection* (MLS2): The “constitutional system” functions as a replicating entity that competes with other systems for adaptive fitness, persisting because it is well-adapted at the societal level?
- (b) *Emergent effects of lower-level selection*: Individual legal memes, doctrines and precedents, compete for transmission and produce systemic rigidity as an aggregate outcome, not as an adaptation?

This article defends (b), but with a qualification that blocks naïve reductionism. Following Okasha (2024), I distinguish between an *organizational hierarchy*, comprising levels of complexity with emergent properties (real, observable), and a *selection hierarchy*, comprising levels where Darwinian adaptation operates (fictitious in law, except at the memetic level). The distinction is not pedantic. It carries predictive consequences that I formalize quantitatively and test empirically.

1.2 Contributions of this Article

The article offers four contributions, each building on the previous.

First, I provide an epistemological critique of MLS2 applied to law. Section 2 demonstrates that treating constitutional systems as Darwinian individuals reintroduces teleology without specifiable causal mechanisms. The critique rests on four objections: absence of societal replication, undefined group fitness, lack of reproductive isolation, and failure of Dennett’s *cui bono* test.

Second, I reconcile Gould and Dawkins. Section 3 accepts Gould’s organizational hierarchy while rejecting his interpretation of it as a selection hierarchy. Macroscopic processes (constitutional lock-in, foundational punctuations) are emergent effects of memetic selection, not group-level adaptations. I identify three distinct causal modes in legal evolution: memetic gradualism, coordinative emergence, and exogenous punctuation.

Third, Section 4 develops the Constitutional Lock-in Index 3.0 (CLI 3.0), which operationalizes downward constraints through the geometry of the Birkhoff polytope. The index decomposes institutional rigidity into three measurable dimensions (spatial, temporal, social) and generates falsifiable predictions about reform feasibility.

Fourth, Section 5 presents preliminary empirical validation. A pilot sample of six Latin American countries (68 reform attempts) yields 78% classification accuracy using CLI 3.0 as the sole predictor, with AUC-ROC of 0.84. Three falsifiable predictions for future validation are stated explicitly.

1.3 Location in the Literature

The work engages three currents simultaneously, which accounts for both its ambition and its vulnerability.

Cultural evolution and memetics. Richerson and Boyd (2005), Mesoudi (2011), and Henrich (2016) apply Darwinian frameworks to culture but rarely distinguish MLS1 (a bookkeeping device that tracks how selection at various levels affects gene frequencies) from MLS2 (a substantive claim that groups function as Darwinian individuals). This confusion generates apparently heated debates that dissolve once the distinction is made explicit. I adopt the critique of group selectionism in culture advanced by Pinker (2012), who identified “the false allure of group selection” in social science, and by Tooby and Cosmides (2023), who extended that critique to recent applications.

Law as extended phenotype. My previous work (Lerer, 2025a, 2025b) establishes game-theoretic microfoundations for legal norms as evolutionarily stable strategies (ESS). Those papers operate entirely within memetic selection, providing the “Micro” foundation. This article extends that framework by incorporating higher-level effects, the “Macro” dimension, without abandoning single-level memetic selection. The integration is possible

precisely because I adopt Okasha's constraint interpretation rather than Gould's selection interpretation of macroscopic patterns.

Philosophy of biology. Okasha (2006, 2024), Godfrey-Smith (2009), Sober and Wilson (1998), and Williams (1966) have debated selection units since the 1960s. This article translates those insights into comparative institutional analysis. The translation is not merely illustrative. I show that different positions in the philosophical debate generate different quantitative predictions about institutional persistence, and that those predictions are empirically discriminable.

2. Rejecting Multilevel Selection Type 2 (MLS2) in Law

2.1 The MLS1 versus MLS2 Distinction

Samir Okasha (2024), in his updated entry for the *Stanford Encyclopedia of Philosophy*, provides the clearest available formulation of a distinction that has generated decades of confusion.

MLS1 (Multilevel Selection Type 1) is an accounting framework. It tracks how selection operating at various levels affects replicator (gene or meme) frequencies. Formally, it decomposes the total change in gene frequency into between-group and within-group components, following the Price equation. MLS1 does not postulate higher-level entities as replicators. It is mathematically equivalent to gene-level selection with contextual effects. Kin selection, for example, can be reformulated as individual selection in the context of population structure, and the two formulations are provably identical (Hamilton, 1964; Price, 1970).

MLS2 (Multilevel Selection Type 2) makes a far stronger claim. It treats groups, species, or other higher-level entities as Darwinian individuals in their own right: entities that replicate, vary, and transmit characteristics to "offspring groups." MLS2 is incompatible with strict gene-centrism because it posits irreducible group-level adaptations, traits that evolved *because* they benefit the group, through a process of differential group reproduction. For MLS2 to apply, four conditions must hold: (i) groups must replicate (generate descendant groups); (ii) group characteristics must be heritable (transmitted vertically from parent to offspring groups); (iii) groups must vary in their emergent properties; and (iv) some groups must have higher fitness than others, where fitness is measured by differential group reproduction.

The perennial confusion is that MLS1 is frequently presented as if it were MLS2. An author reports that groups differ in some property (true, uncontroversial), shows that group membership affects individual outcomes (MLS1, also uncontroversial), and then concludes

that group selection is operating (MLS2, a non sequitur). This slippage, which Pinker (2012) identified as endemic in cultural evolution literature, generates spurious debate between positions that are not actually in disagreement once the distinction is enforced.

2.2 Four Fatal Objections to MLS2 in Legal Systems

Objection 1: Absence of societal replication

Societies do not reproduce. When Argentina persists from 1983 to 2025 under the same labor regime, it is not “generating daughter societies” with inherited constitutional characteristics. Persistence is the *stability of an equilibrium*, not Darwinian replication. A society that maintains its institutions for forty years is more like a crystal that retains its structure than like an organism that produces offspring.

The contrast with biology is instructive. A species that persists for ten million years does generate descendant populations through speciation: geographic isolation, genetic drift, and differential selection in distinct environments produce new species that inherit (with variation) the ancestral genome. Here the analogy to MLS2 has purchase, though Gould overstated its adaptive importance. But legal systems do not undergo speciation. Argentina did not bud off “daughter Argentinas” in 1994 when it reformed its constitution. It modified itself. This is more akin to somatic mutation (non-heritable within the Darwinian framework) than to germline reproduction.

One might argue that colonial law represents a form of “legal speciation”: Spain’s legal system generated “descendant” systems across Latin America. But the mechanism is memetic transmission (individual doctrines, codes, and institutional templates migrating to new environments), not group reproduction. Spain did not reproduce as a society; specific legal memes spread through colonial administration and persisted differentially depending on local environmental compatibility. This is MLS1, not MLS2.

Objection 2: Undefined group fitness

For MLS2 to operate, group fitness must be definable independently of the fitness of constituent replicators. What does it mean for a “constitutional system” to have high fitness?

Typical proposals collapse on examination. *Higher GDP per capita* confuses economic output with reproductive aptitude. GDP is an outcome of many factors, of which institutional design is one; treating it as fitness inverts the explanatory structure by conflating explanandum with explanans. *Greater political stability* is circular: stability simply is the persistence we seek to explain. Defining fitness as persistence and then explaining persistence by fitness is not a theory; it is a tautology. *Greater adoption by other jurisdictions*

does involve genuine transmission, but transmission of *legal memes* (the lower level), not of “societies.” When Chile adopted a proportionality test inspired by German constitutional law, Germany did not reproduce; a specific doctrine traveled to a new institutional environment.

No metric of societal fitness exists independently of the success of constituent memes. This is not a temporary empirical limitation that additional data might resolve. It is a conceptual incoherence: the very concept of “societal fitness” in the Darwinian sense presupposes a population of competing societies undergoing differential reproduction, which is precisely what does not exist.

Objection 3: No reproductive isolation

In biology, species are reproductively isolated; no gene flow crosses species boundaries (the Biological Species Concept). This isolation is a necessary condition for MLS2 because without it, group-level traits cannot be maintained against the eroding force of migration. In law, constitutional systems *exchange doctrines constantly*. Three examples illustrate the scale of the phenomenon.

Judicial review spread from *Marbury v. Madison* (United States, 1803) to over 150 jurisdictions across both common law and civil law traditions within two centuries. The proportionality test (*Verhältnismäßigkeit*) traveled from 1950s German constitutional law across the globe via constitutional tribunals, becoming perhaps the most widely adopted analytical framework in comparative constitutional adjudication. Due process migrated from medieval English common law (“*per legem terrae*” in Magna Carta, 1215) into modern civil law systems that had no prior tradition of it. This massive horizontal doctrinal flow destroys the population structure that MLS2 requires. If genes flowed freely between species at this rate, speciation would be impossible and the species concept itself would dissolve.

Objection 4: Dennett’s *cui bono* test

Daniel Dennett (1995) proposed a diagnostic question for evaluating adaptationist claims: *cui bono*? For whose benefit does the trait exist? If the answer is “the group,” the theorist must specify the causal mechanism that connects group benefit to increased frequency of the trait in question. Without such a mechanism, the claim is teleological: it identifies a beneficiary without explaining the causal pathway.

Applied to Argentina’s Article 14bis: the provision persists despite 47.3% labor informality, 23.7% youth unemployment, and fiscal costs estimated at 2.3% of GDP. The MLS2 answer (“it benefits Argentine society by protecting labor rights”) is empirically false. The regime does not protect the 47.3% of workers in the informal sector, who enjoy none of its benefits

while bearing its costs through reduced formal employment opportunities. The memetic answer (“it benefits carriers of specific memes, namely unions with institutional veto power, labor court judges whose jurisdiction depends on doctrinal complexity, and labor law faculties whose curricular relevance requires the doctrine’s persistence”) is empirically true. These three groups of meme carriers have successfully blocked 23 consecutive reform attempts between 1983 and 2025.

The *cui bono* test reveals that MLS2 inverts causality. Norms do not persist because they benefit the group. Groups form *because sustaining norms* increases their members’ memetic fitness. The union that defends ultra-activity does not persist because ultra-activity benefits society; the union’s institutional survival *depends on* ultra-activity’s persistence, which gives its leaders organizational relevance, membership dues, and political leverage.

2.3 Illustrative Case: Argentina versus Uruguay in Labor Regulation

The Argentina-Uruguay comparison provides a natural experiment because the two countries share linguistic, cultural, and legal heritage (both belong to the civil law tradition, both received strong influence from European labor law, both experienced similar political cycles of democracy and authoritarianism in the twentieth century) while diverging sharply in institutional architecture.

Indicator	Argentina	Uruguay
Constitutional anchor	Art. 14bis (1949/1957)	Ordinary legislation, no constitutional lock
CLI 3.0	0.87 (extreme, T-concentrated)	0.31 (balanced)
Ultra-activity	Indefinite	Limited (2–3 years)
Reform attempts (1983–2025)	23	11
Reform success rate	0%	63% (7/11)
Informality rate	47.3%	24.1%
Youth unemployment	23.7%	14.8%

Table 1. Argentina versus Uruguay: parallel histories, divergent institutional outcomes.

The MLS2 prediction is straightforward: Argentina, with worse labor market outcomes by every available metric, should collapse from “systemic maladaptation”; Uruguay, with superior outcomes, should persist as a “superior system.” The observation is different. Argentina persists *without reform* through memetic lock-in, not group fitness. Uruguay

reforms *successfully* because it lacks lock-in, not because it possesses an inherently “superior system.” Both systems are stable; they are stable for different reasons, and those reasons operate at the memetic level.

MLS2 does not explain differential persistence. It inverts cause and effect. Lock-in is not adaptation. It is institutional inertia generated by the strategic alignment of meme carriers within an evolutionarily stable strategy (ESS) in the constitutional game. The system that produces 47.3% informality is not “adapted”; its constituent memes are well-adapted to the environment that sustains them, which is a different statement entirely.

3. Accepting Emergent Effects: The Gould-Okasha Causal Hierarchy

3.1 Where Gould Was Right (Partially)

Having rejected MLS2 does not require rejecting everything Gould observed. Gould (2002) correctly identified three macroevolutionary phenomena that remain valid regardless of one’s position on selection units.

Speciation processes are not reducible to gradual microevolution. Gould and Eldredge’s (1972) punctuated equilibria model documented that new species typically appear geologically “instantaneously” (over 5,000 to 50,000 years, fast by geological standards) rather than through continuous gradual change. The fossil record supports this pattern in most lineages, even if the mechanism (peripatric speciation, founder effects) operates through individual-level selection in small, peripherally isolated populations.

Mass extinctions operate as environmental filters on entire clades. The K-T boundary event (asteroid impact, approximately 66 million years ago) eliminated roughly 75% of species, including all non-avian dinosaurs. The Permian-Triassic extinction (approximately 252 million years ago) eliminated approximately 96% of marine species. These events are not microevolutionary; no population genetics equation predicts when an asteroid will strike. They require geology, climatology, and ecosystem dynamics for explanation.

Adaptive radiations produce explosions of diversity that are difficult to explain by gradual accumulation alone. The Cambrian radiation generated over 20 new phyla in approximately 10 million years, a pace without precedent before or since. These events suggest that the fitness landscape itself transforms during certain periods, creating new ecological niches faster than organisms can fill them.

Gould’s *error* lay in interpreting these phenomena as evidence of *species selection*: treating species as adaptive individuals whose differential reproduction explains macroevolutionary patterns. Okasha (2024) and others have demonstrated that all three phenomena can be

explained through selection at lower levels (genes and organisms) combined with higher-level constraints, drift in structured populations, and stochastic sampling effects at the macroscale. Species do not need to be selection units for macroevolutionary patterns to exist.

What must be retained from Gould is the *phenomenological irreducibility* of the macroscopic. The fact that all causation ultimately involves lower-level entities does not mean that macroscopic descriptions are eliminable from scientific explanation. Mass extinctions cannot be predicted, even in principle, from population genetics models alone; they require independent macroscopic variables (asteroid trajectories, volcanic activity, atmospheric chemistry). The irreducibility is epistemic and predictive, even if it is not ontological in the strong sense Gould intended.

3.2 Downward Causation by Constraint (Okasha)

Okasha (2024) formalizes how higher levels affect lower levels without being selection units themselves. The concept is *top-down causation by constraint*.

Definition: A level $N+1$ exercises downward causation on level N if the emergent properties of $N+1$ restrict the space of possibilities accessible to entities at N , without $N+1$ possessing a fitness function of its own.

The K-T extinction provides the paradigmatic example. The asteroid impact (a cosmic-geological event at the higher level) eliminated non-avian dinosaurs. This constitutes downward causation: the impact restricted which phenotypes were viable (cold tolerance, small body size, capacity for hibernation or torpor) without “selecting species.” There is no “selection of cold-tolerant species.” There is selection of individuals in a suddenly transformed environment. The transformation is a higher-level event; the selection remains at the lower level. Okasha’s framework separates these two claims, which Gould had run together.

The distinction matters because it determines what predictions we make. If species are selection units, we should expect species-level adaptations, traits that exist because they benefited the species. If constraints are the correct interpretation, we should expect traits that survived environmental filters without having been designed for that purpose. The empirical record overwhelmingly supports the latter: mammalian traits that proved advantageous after the K-T event (endothermy, small body size, burrowing habits) evolved for other reasons long before the asteroid.

3.3 Application to Law: Lock-in as Downward Constraint

Translating Okasha’s framework into the legal-memetic domain:

Biological Level (Gould)	Legal-Memetic Level (CLI 3.0)	Emergent Process	Selection?
Gene	Legal meme (doctrine, precedent)	Differential replication	✓ Yes
Organism	Individual legal strategy	Phenotypic expression	X No (vehicle)
Population/Deme	Institution (court, bar association)	Strategic coordination	X No (ESS)
Species	Constitutional system	Lock-in (CLI > 0.75)	X No (constraint)
Clade	Legal family (Common/Civil Law)	Path dependence	X No (contingency)

Table 2. Mapping between biological and legal-memetic hierarchical levels. Selection operates only at the memetic (gene) level.

Argentina's Article 14bis illustrates the complete mechanism in four steps.

Step 1: Memetic competition (lower level). Specific doctrines compete for transmission. Meme A ("indefinite ultra-activity protects workers") competes with Meme B ("limited ultra-activity allows sectoral adaptation"). Meme A replicates more successfully because it generates precedents that cite prior precedents (positive judicial feedback loop), creates organizations institutionally dependent on its persistence (unions with veto power derived from ultra-activity), and enters law school curricula (generational transmission via education). Each of these mechanisms is a standard memetic replication channel.

Step 2: Emergent accumulation. Sixty-eight years of Meme A's replicative success produces an emergent property at the system level: a CLI 3.0 of approximately 0.87, concentrated almost entirely in the temporal dimension ($T \approx 0.87$).

Step 3: Downward constraint. The elevated CLI restricts which future memes can enter the system. Meme B (flexibilization) requires constitutional amendment, an option whose political cost is prohibitive given the institutional architecture. Incremental alternatives (sectoral exceptions, regulatory relaxation) are blocked by doctrinal coherence requirements: admitting one exception would undermine the logical structure of 68 years of precedent. The system acts as a filter, admitting only memes compatible with indefinite ultra-activity.

Step 4: Reinforcement. The filter effect strengthens the very memes that produced it. Each blocked reform attempt validates the doctrine's persistence ("the system withstood another challenge"), providing fresh precedential material and reinforcing the reputational

costs of innovation. The system enters a positive feedback loop that transforms lock-in from reversible equilibrium to deep-basin attractor.

The CLI does not “compete” with other systems. It has no fitness. It is an *emergent property* that transforms the selective landscape for lower-level memes. This is Okasha’s downward causation by constraint, applied to the institutional domain.

3.4 Three Causal Modes in Legal Evolution

Integrating Gould, Okasha, and memetic theory yields three distinct causal modes that exhaust the evolutionary dynamics of legal systems.

Mode 1: Memetic gradualism (analogous to microevolution). Doctrines evolve incrementally through selection on transmission fidelity, transmission scope, and environmental compatibility (the TCI parameters in the lusMorfos framework). Due process in common law has undergone approximately 800 years of gradual refinement since Magna Carta (1215), with each generation of jurists inheriting, modifying, and retransmitting the doctrine. The pace is slow, the direction is not predetermined, and the outcomes depend on local selective pressures. This mode accounts for the majority of everyday legal change.

Mode 2: Coordinative emergence (stable macroevolution). Institutional equilibria (ESS) form at higher levels without being objects of direct selection. The US Electoral College (CLI 2.0 $\approx$ 0.800) persists as a Nash equilibrium in the federal game: no individual state benefits from unilateral withdrawal, and no coalition of states has sufficient incentive to coordinate reform. The equilibrium is *stable*, but it was not *selected for*; it emerged from the strategic interaction of lower-level agents and, once established, resists perturbation through its own structural properties. This is the constitutional equivalent of species stasis in Gould’s punctuated equilibria model, but without species selection.

Mode 3: Exogenous punctuations (revolutions). Systemic reconfiguration driven by non-legal events: wars, severe economic crises, revolutions, collapses of political regimes. India’s Constitution of 1949 followed independence from British colonial rule; South Africa’s of 1994 followed the end of apartheid; Germany’s Basic Law of 1949 followed military defeat and occupation. These punctuations are analogous to mass extinctions: they clear the institutional landscape, creating ecological opportunities for novel legal memes that would have been unviable under the prior regime. The Constituent Assembly that drafted India’s constitution could introduce provisions (universal suffrage, affirmative action, directive principles) that would have been inconceivable within the colonial legal framework.

Only Mode 1 involves selection (memetic). Modes 2 and 3 are emergent effects and external constraints, respectively. Gould conflated Modes 2 and 3 with “species selection.” This framework distinguishes them as downward causation by constraint (Mode 2) and

exogenous perturbation (Mode 3). The taxonomy is not merely classificatory; it generates different predictions. Mode 1 changes are incremental and reversible. Mode 2 equilibria resist perturbation but can be broken by sufficiently large shocks. Mode 3 events are unpredictable from within the legal system and create *windows of opportunity* whose timing and duration determine which reforms are feasible.

4. Quantitative Formalization: CLI 3.0 as a Geometry of Constraints

4.1 Motivation: From Metaphor to Metric

The preceding sections establish three propositions: constitutional lock-in is real (an observable emergent effect), lock-in is not a selection unit (it possesses no fitness of its own), and lock-in operates as a downward constraint on future memetic evolution. What remains is operationalization. How do we measure lock-in intensity? When does a system cross the point of no return? What distinguishes a reversible equilibrium from an irreversible attractor?

The Constitutional Lock-in Index 3.0 answers these questions using the formal mathematics of constraints: lattice theory and linear algebra. *Methodological note:* The complete mathematical formalization, including dynamic equations, sensitivity analysis, and Bayesian estimation procedures, is presented in a separate methodological paper (Lerer, 2026, in preparation). This section summarizes key concepts for the non-mathematical reader while preserving enough formal structure to enable replication of the empirical results.

4.2 Three Dimensions of Institutional Lock-in

CLI 3.0 decomposes institutional rigidity into three observable components, each capturing a distinct mechanism by which legal memes entrench themselves.

Dimension S (Spatial): horizontal diffusion across jurisdictions. How many countries or sub-national jurisdictions have adopted doctrine X? Judicial review, adopted by over 150 jurisdictions, scores $S \approx 0.85$. Argentina's amparo, which remains jurisdictionally specific, scores $S \approx 0.15$. Spatial diffusion matters because it creates an international support network for the doctrine: judges in multiple countries cite each other's precedents, reinforcing the doctrine's perceived legitimacy and making local reform appear eccentric or regressive.

Dimension T (Temporal): vertical persistence across generations. How many generations of jurists have sustained doctrine X? Article 14bis of Argentina's constitution (68 years, three or more complete jurist generations) scores $T \approx 0.87$. Macri's 2016 labor reform (seven years,

less than one generation) scores $T \approx 0.22$. Temporal persistence matters because it creates path dependence through accumulated precedent: each generation inherits not just the doctrine but the entire interpretive apparatus built around it, making the cost of departure grow exponentially with each additional generation.

Dimension Soc (Social): penetration across heterogeneous organized sectors. How many organized sectors depend institutionally on doctrine X? Argentina's pension regime, which binds unions, universities, obras sociales (union-administered health insurance), and provincial pension funds, scores $Soc \approx 0.78$. A sector-specific subsidy affecting only the metalworkers' union scores $Soc \approx 0.25$. Social penetration matters because it creates a distributed veto system: reform requires simultaneous consent (or defeat) of multiple organized actors, each with different interests but a shared stake in the doctrine's preservation.

The mathematical constraint $S + T + Soc = 1.0$ (normalization) reflects a non-trivial empirical regularity: any legal system consumes *finite institutional resources*. These resources include space in jurisprudence (bounded by judicial cognitive capacity), time in law school curricula (typically 24 months of specialization, partitioned across doctrines), and legislative attention (the parliamentary agenda is a zero-sum allocation problem). Increasing lock-in in one dimension *redistributes* resources from the others. A doctrine that achieves extreme temporal persistence ($T \rightarrow 1$) does so at the expense of spatial diffusion and social penetration, because institutional resources that maintain generational transmission are diverted from horizontal diffusion and cross-sectoral extension.

4.3 The Birkhoff Polytope: Geometry of Possible Configurations

The three dimensions under the constraint $S + T + Soc = 1$ define a simplex: a triangle in three-dimensional space. Any legal system occupies a point within this triangle. The vertices represent extreme configurations:

Vertex $S = (1, 0, 0)$: purely spatial lock-in. A recent international treaty with no generational depth and no organizational dependence. Highly diffused but shallow; vulnerable to unilateral withdrawal. Vertex $T = (0, 1, 0)$: purely temporal lock-in. An ancient doctrine in a single jurisdiction maintained by a narrow elite. Deep but fragile; vulnerable to jurisdictional reform if the elite loses power. Vertex $Soc = (0, 0, 1)$: purely social lock-in. A new norm adopted simultaneously by many sectors without historical depth. Broad but unstable; vulnerable to coordinated defection if sectors find alternative institutional arrangements. The center $(0.33, 0.33, 0.33)$: perfect balance, with lock-in distributed equitably across dimensions.

The geometric insight is that lock-in is *not* the absolute magnitude of S, T, and Soc (that would merely count institutional presence). Lock-in is the *deviation from the center*. A system with $S = T = \text{Soc} = 0.33$ exhibits low lock-in despite institutional presence across all three dimensions, because it is not concentrated in any single dimension and therefore lacks the self-reinforcing dynamics that concentration produces.

The simplified CLI 3.0 formula is: $\text{CLI}^3 = 1 - \text{Balance}(S, T, \text{Soc})$, where Balance measures the system's proximity to the centroid. Systems near the center score low on CLI^3 ; systems near a vertex score high.

4.4 Empirical Interpretation: Critical Threshold at $\text{CLI}_3 \approx 0.75$

Preliminary validation with six Latin American countries (1983–2025):

Country	CLI 3.0	S	T	Soc	Reforms	Informality
Uruguay	0.31	0.29	0.35	0.36	63% (7/11)	24.1%
Chile	0.24	0.42	0.28	0.30	83% (5/6)	28.4%
Brazil	0.52	0.38	0.35	0.27	54% (6/11)	39.6%
Colombia	0.58	0.25	0.48	0.27	38% (3/8)	48.7%
Mexico	0.68	0.18	0.52	0.30	22% (2/9)	56.9%
Argentina	0.87	0.12	0.87	0.01	0% (0/23)	47.3%

Table 3. CLI 3.0 scores and reform outcomes for six Latin American countries (1983–2025).

The observable pattern admits three zones. Below $\text{CLI}_3 = 0.40$: high reformability (above 60% success). Between 0.40 and 0.70: intermittent reform (30–55% success). At or above 0.75: practical irreformability (below 5% success). The correlation coefficient is $r(\text{CLI}_3, \text{reform rate}) = -0.94, p < 0.01$.

The causal hypothesis (not yet demonstrated) for the critical threshold: above $\text{CLI}_3 \approx 0.75$, the system enters a positive feedback regime. High T generates more precedents, which increases T further. Additional precedents create complex jurisprudence demanding specialized organizations (law firms, academic departments), raising Soc. Those organizations have strategic incentives to preserve complexity, because their institutional relevance depends on it, so they block simplification. Successful blocking reinforces T and Soc. The cycle transforms lock-in from a reversible stable equilibrium (an equilibrium from which the system can be dislodged with moderate effort) to a deep-basin attractor (one requiring massive exogenous shock for displacement).

4.5 Connection with Okasha: CLI_3 as a Constraint Operator

Formally (without mathematical heavy lifting), CLI_3 acts as a filter on the space of possible legal memes.

Memetic space without constraint: all memes compatible with legal natural language can compete. In a jurisdiction without binding precedent, any interpretation of “due process” is viable. The selective landscape is flat, with multiple peaks of comparable height.

Memetic space under $CLI_3 \approx 0.87$: only memes coherent with 68 years of prior doctrine are viable. Novel memes contradicting established precedent face reputational costs (judges who innovate are criticized as “activists” and risk reversal on appeal), procedural instability (resolving case X differently jeopardizes 1,000 similar pending cases whose disposition assumed the established interpretation), and coordination impossibility (changing doctrine X requires simultaneously changing doctrines Y, Z, and W for coherence, a coordination problem that no single actor can solve). The selective landscape has been reshaped into a configuration with a single very tall, steep peak (the indefinite ultra-activity meme) surrounded by deep valleys that make transition to any alternative peak prohibitively costly.

CLI_3 is not “fitness of the Argentine system.” It is the *topology of the selective landscape* that confronts any new meme. Argentina has a single-peak landscape. Uruguay has a multi-peak landscape. The difference is not in the memes themselves (both countries have access to the same doctrinal repertoire) but in the topological structure of the environment in which those memes compete.

5. Empirical Validation and Predictive Power

5.1 Regime Discrimination (Pilot Analysis, $n = 6$)

The hypothesis under test is that $CLI_3 > 0.75$ predicts constitutional reform failure with at least 75% accuracy. The data comprise 68 reform attempts across six countries: 23 in Argentina (1983–2025), 11 in Uruguay, 6 in Chile, 8 in Colombia, 11 in Brazil, and 9 in Mexico. The method is binary classification (success/failure) using CLI_3 as the sole predictor, with the optimal threshold estimated at $CLI_3^* = 0.73$ (maximizing the F1-score).

Metric	Value	95% CI
Overall accuracy	78.2%	[69.1%, 85.3%]
Sensitivity (detects failures)	87.5%	[75.0%, 94.1%]
Specificity (detects successes)	72.3%	[60.8%, 81.7%]
AUC-ROC	0.84	[0.76, 0.91]

Table 4. Classification performance of CLI_3 as sole predictor of reform outcome ($n = 68$ attempts).

The null model (prediction by regional base rate) achieves 56.3% accuracy. CLI_3 provides a gain of 21.9 percentage points ($p < 0.001$, permutation test with 10,000 iterations). The model is notably stronger at detecting failures (sensitivity 87.5%) than at detecting successes (specificity 72.3%), which is expected: lock-in predicts rigidity more reliably than low lock-in predicts flexibility, since low-lock-in systems can still fail reforms for reasons unrelated to institutional structure (political timing, economic context, coalition dynamics).

Analysis of false positives

Two false positives deserve examination. Brazil 2017 ($CLI_3 = 0.52$, model predicted failure at 68% confidence, observed outcome: success) and Colombia 2015 ($CLI_3 = 0.58$, predicted failure at 73%, observed: success). Both cases involve *exogenous shocks* that temporarily reduced the political cost of reform. Brazil experienced a severe fiscal crisis during 2015–2016, with GDP contracting 3.5% in 2015 and 3.3% in 2016; this generated extraordinary political pressure that enabled the Temer government to build a bipartisan reform coalition. Colombia was in the midst of the FARC peace process (2012–2016), which created a window of institutional fluidity in which reforms that would otherwise face blocking coalitions became temporarily feasible.

This analysis suggests a model extension: $CLI_{3_effective} = CLI_{3_static} \times (1 - \text{shock intensity})$, where shock intensity is a scalar between 0 and 1 measuring the severity of the exogenous perturbation. When shock = 0 (business as usual), $CLI_{3_effective}$ equals CLI_{3_static} . When shock approaches 1 (regime-level crisis), $CLI_{3_effective}$ approaches 0, meaning that even high-lock-in systems become temporarily reformable. The challenge is operationalizing “shock intensity” in a non-circular way; GDP contraction, political crisis indices, and institutional fluidity measures are candidates, but the specification requires future work.

5.2 Cross-Jurisdictional Validation

Is the threshold $CLI_3^* \approx 0.75$ specific to Latin America, or does it generalize?

Preliminary European data are consistent. Spain ($CLI_3 \approx 0.48$): seven labor reforms between 1978 and 2023, four successful (57%). Italy ($CLI_3 \approx 0.67$): nine reforms between 1990 and 2023, three successful (33%). Germany ($CLI_3 \approx 0.55$): twelve reforms between 1990 and 2023, six successful (50%). The European threshold appears similar at $CLI_3^* \approx 0.70$ – 0.75 . Italy, near the threshold, shows high variance consistent with the transition zone predicted by the model (systems near the threshold should oscillate between reform success and failure depending on the magnitude of contemporaneous shocks).

The European sample remains small ($n = 28$ attempts across three countries). Expansion to 60 transitions (20 countries across three periods, 1985–2025) is underway. Preliminary

analysis of additional Latin American cases (Peru, Ecuador, Bolivia, Paraguay) is consistent with the pattern but requires formal coding of reform attempts before inclusion.

5.3 Falsifiable Predictions for Future Validation

Prediction 1 (Post-threshold irreversibility): If $CLI_3(t_0) > 0.75$ and no shock η exceeding η_{crit} occurs (where η_{crit} denotes a constitutional crisis on the scale of revolution, military defeat, or state collapse), then $CLI_3(t_0 + 10 \text{ years}) \geq CLI_3(t_0)$. The prediction is that lock-in above the threshold is self-reinforcing absent massive exogenous perturbation. Test: track Mexico ($CLI_3 = 0.68$, near the threshold) from 2025 to 2035. If Mexico crosses 0.75 during this period (plausible given current MORENA institutional consolidation), it should become irreformable in the labor domain within a decade. If it remains below 0.75, reforms should remain feasible, albeit difficult.

Prediction 2 (Goldilocks zone): Systems with $0.40 < CLI_3 < 0.65$ should exhibit greater macroeconomic stability (lower GDP volatility, lower structural unemployment) than systems below 0.30 (hyperflexibility, which produces policy instability and regulatory uncertainty) or above 0.75 (rigidity, which produces informality and fiscal drain). The prediction rests on the assumption that institutional design has a non-monotonic relationship with economic performance: too little lock-in prevents credible commitment, too much lock-in prevents adaptation. Test: panel regression of 60 countries controlling for per capita GDP, political institutions, and resource endowment.

Prediction 3 (Reform-counterreform cycles): Systems with $0.55 < CLI_3 < 0.70$ should exhibit periodic oscillations (10–15-year cycles) between flexibilization attempts and rigidity restoration. The mechanism is that systems in this zone are close enough to the threshold for occasional shocks to push them above it (triggering lock-in dynamics), but far enough below for those dynamics to be incomplete, allowing the next shock to push them back (triggering counterreform). Brazil and Colombia, both in the intermediate zone, should exhibit this oscillatory pattern. Test: spectral analysis (Fourier transform) of reform time series in both countries, looking for periodicities in the 10–15-year range.

6. Synthesis: Reconciling Gould with Dawkins via Okasha

6.1 Theoretical Coherence Matrix

Proposition	Dawkins	Gould	Okasha	CLI 3.0
Sole selection unit: genes/memes	✓	✗	✓	✓

Irreducible macroscopic processes exist	X	✓	✓	✓
Higher levels possess their own fitness	X	✓	X	X
Downward causation by constraint	?	?	✓	✓
Operational mathematical formalization	Verbal	Verbal	Philosophical	Quantitative

Table 5. Theoretical coherence matrix. CLI 3.0 materializes Okasha's synthesis: it accepts Gould's organizational hierarchy without accepting MLS2.

6.2 Implications for Evolutionary Legal Theory

First: the end of the “system selection” metaphor. Expressions such as “the Argentine system was selected for...” are category mistakes. Systems persist through memetic inertia (ESS in the constitutional game), not through adaptive group fitness. This terminological discipline is not pedantic; it prevents teleological explanations from contaminating institutional analysis. When a political scientist says “Argentina’s labor regime persists because it serves a social function,” she commits the same error as a biologist who says “the peacock’s tail exists because it benefits the species.” The peacock’s tail benefits genes for long tails; the labor regime benefits memes for indefinite ultra-activity. The fact that these beneficiaries are not the same as the putative group-level beneficiaries is the entire point.

Second: institutional design as constraint engineering. If lock-in is geometry, it becomes designable. Constitutions can be drafted with a target CLI_3 (for instance, 0.50 ± 0.10), incorporating automatic sunset clauses that reduce T by forcing periodic doctrinal renewal, periodic revision mechanisms that prevent dimensional concentration, and cross-jurisdictional review procedures that increase S without increasing T. The design problem becomes: what configuration (S, T, Soc) maximizes institutional stability (credible commitment to long-term rules) subject to a minimum reformability constraint (the system must remain capable of adaptation when circumstances change)?

Third: prediction of reform windows. Exogenous shocks (economic crises, generational turnover, regime transitions) temporarily reduce effective CLI_3 by disrupting the positive feedback loops that sustain lock-in. The timing of reform attempts becomes an optimization problem under temporal constraints: the window opened by a shock closes as lock-in dynamics reassert themselves. Modeling this requires dynamic $CLI_3(t)$ equations (that will

be object of a separate methodological paper), but the qualitative implication is already actionable. Argentina's 2001–2002 crisis was a shock of magnitude approximately $\eta \approx 0.7$ (GDP contraction of 10.9%, institutional collapse, five presidents in ten days). If $CLI_3_effective = 0.87 \times (1 - 0.7) = 0.26$, the system was temporarily in the reformable zone. The Duhalde and Kirchner governments chose to restore the prior equilibrium rather than exploit the window; this was a political decision, not a structural inevitability.

6.3 Connection with Khurshid's MRSP/MSSP Framework

Professor Showan Khurshid's¹ (2006, 2025) Tgmenks framework provides a political-philosophical complement to the quantitative analysis developed here. Khurshid distinguishes two modes by which political arrangements address the struggle for power. *MRSP (Method of Regulating the Struggle for Power)* systems structure competition under rules that bind all participants, including the most powerful. *MSSP (Method of Suppressing the Struggle for Power)* systems construct mechanisms ensuring that only one faction can effectively exist.

Khurshid's core insight is that political belief systems (PBS) serve either function. In MRSP systems, PBS provide the normative framework within which power competition occurs; they constrain the means available to competitors without eliminating competition itself. In MSSP systems, PBS are deployed to sanctify the dominant faction's power, to identify challengers as morally illegitimate, and to apply moral sanctions asymmetrically, with severity toward outsiders and impunity toward insiders.

The MRSP/MSSP distinction does not map neatly onto democracy versus authoritarianism. A formal democracy can operate as MSSP if its institutional structure ensures that only one political tendency can realistically govern (through gerrymandering, judicial capture, or media control). A non-democratic system can exhibit MRSP features if power competition occurs within institutional channels, even if those channels exclude popular participation.

I propose, as an extension that is not present in Khurshid's original formulation, that the CLI_3 threshold of approximately 0.75 corresponds to the MRSP→MSSP transition in institutional behavior. Below 0.75, the system preserves regulatory space: competing doctrines can challenge established norms, reform is politically feasible, and sanctions for innovation are roughly symmetric. Above 0.75, the system suppresses regulatory space: the dominant

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doctrinal meme has captured the institutional infrastructure, challengers face asymmetric sanctions (reputational costs for judges, procedural impossibility for legislators), and the system reproduces the dominance of the incumbent meme regardless of its adaptive value for the population at large.

If this extension survives empirical testing, it would connect CLI 3.0's institutional geometry with Khurshid's political-philosophical framework, providing a quantitative criterion for what Khurshid identifies through qualitative analysis. The MRSP/MSSP boundary would become a measurable feature of the constraint landscape rather than a purely conceptual distinction. The threshold $CLI_3 \approx 0.75$ would acquire not only predictive value (discriminating reformable from irreformable systems) but also normative significance (marking the boundary between institutional arrangements that preserve the possibility of democratic self-correction and those that have effectively foreclosed it).

6.4 Contribution to the Philosophy of Biology Debate

This article demonstrates, perhaps unexpectedly, that comparative law can contribute to resolving a debate in the philosophy of biology that has persisted for fifty years. The demonstration rests on showing that the competing positions generate different quantitative predictions in the legal domain, and that those predictions are empirically discriminable.

If Gould were right about MLS2: Argentina should collapse from systemic maladaptation; its labor regime produces worse outcomes than Uruguay's on every available metric (not observed, since Argentina persists without collapse). "Optimal" systems should converge globally as superior systems outcompete inferior ones (not observed, given the stable diversity of constitutional regimes). Group fitness should predict institutional persistence: $r(\text{GDP, persistence}) \approx 0.12$ (insignificant in the pilot sample).

If the Dawkins-Okasha position is correct: Persistence should depend on memetic ESS rather than group fitness (observed: CLI_3 predicts with 78% accuracy, compared to the null model's 56.3%). Stable diversity of regimes should be the norm, since multiple ESS can coexist in the same environment (observed: Latin American countries with similar cultural backgrounds sustain dramatically different institutional configurations). Lock-in should be formalizable as topological constraint on the selective landscape (achieved through CLI 3.0 and empirically testable).

Comparative law resolves the debate because it provides exactly what the biological literature has difficulty providing: large numbers of well-documented cases where institutional persistence can be measured, reform attempts can be counted, and environmental conditions can be compared across jurisdictions. The biological debate is

hampered by the difficulty of observing species selection in real time over geological timescales. The legal debate operates on human timescales (decades, not millennia) with rich documentary records. Dawkins was right about selection units; Gould was right about the existence of higher causal levels. Okasha provides the conceptual synthesis; CLI 3.0 provides the quantitative operationalization.

7. Conclusions and Research Agenda

7.1 Principal Results

Four results emerge from the analysis. *First*, type 2 multilevel selection applied to law lacks causal microfoundations. Societies do not replicate, have no measurable fitness independent of constituent memes, and are not reproductively isolated. The four objections (absence of replication, undefined fitness, no isolation, failed cui bono test) are individually sufficient and collectively devastating. *Second*, the reconciliation of Gould and Dawkins accepts the organizational hierarchy (real emergent effects observable at macroscopic levels) but rejects the selection hierarchy (groups as adaptive units). The bridge concept is Okasha's downward causation by constraint, which permits higher-level effects without requiring higher-level selection. *Third*, CLI 3.0 formalizes downward constraints through the Birkhoff polytope, generating falsifiable predictions with 78% accuracy in the pilot sample (AUC-ROC = 0.84, gain over null model = 21.9 percentage points). *Fourth*, the threshold $CLI_3 \approx 0.75$ marks the transition from reversible equilibrium to irreversible attractor. Argentina (0.87) lies beyond the event horizon; Uruguay (0.31) occupies a stable, reformable zone.

7.2 Limitations and Necessary Extensions

Limitation 1: Sample size. Six countries and 68 reform attempts constitute a pilot, not a definitive test. The correlation $r = -0.94$ is impressive but derived from small n ; confidence intervals for classification accuracy are correspondingly wide (overall accuracy 95% CI: [69.1%, 85.3%]). Expansion to 60 transitions (20 countries across three periods, 1985–2025) is in progress and will include countries from Latin America, Europe, and Asia.

Limitation 2: Mathematical formalization. The dynamic specification of $CLI_3(t)$ as a stochastic process with shocks, developed in the separate methodological paper, employs differential equations and lattice-theoretic structures that require review by specialists in mathematical physics or dynamical systems theory before they can be considered established. The static CLI_3 presented here is more conservative and less vulnerable to mathematical error, but it sacrifices temporal dynamics.

Limitation 3: Causal ambiguity. The correlation between CLI_3 and reform outcomes is robust, but the direction of causation has not been demonstrated through instrumental variables, regression discontinuity, or quasi-experimental design. It is possible (though I consider it implausible) that reform failure causes high CLI_3 rather than the reverse; establishing causal direction requires either natural experiments or structural estimation.

Extension 1: Integration with Khurshid's MRSP/MSSP framework. The proposed correspondence between $CLI_3 > 0.75$ and the MRSP→MSSP transition requires independent validation. One test: expert classification of countries as MRSP or MSSP by political scientists unfamiliar with CLI_3 , followed by comparison with CLI_3 -based classification. If the two classifications agree substantially ($\kappa > 0.6$), the correspondence has independent support.

Extension 2: Temporal dynamics. Modeling $CLI_3(t)$ as a stochastic process with shocks, identifying parameters through Bayesian MCMC estimation, and predicting country-specific trajectories for 2025–2050. Mexico's trajectory is the most interesting test case: currently at $CLI_3 = 0.68$ (near threshold), its institutional evolution under MORENA will either push it above 0.75 (validating the irreversibility prediction) or stabilize below it (constraining the model's scope).

Extension 3: Optimal constitutional design. Using CLI 3.0 as an objective function: what configuration (S, T, Soc) maximizes institutional stability subject to a minimum reformability constraint? This is a constrained optimization problem on the simplex, amenable to standard techniques. Preliminary analysis suggests that the optimal zone is $CLI_3 \in [0.45, 0.60]$, but this result depends on how “stability” and “reformability” are operationalized.

7.3 Closing Argument

The Gould-Dawkins dispute is not a rhetorical battle between academic personalities. It is a scientific controversy about the causal structure of the world: whether macroscopic patterns require macroscopic selection, or whether they emerge from microscopic selection under macroscopic constraints. This article shows that both were partially right, and that Okasha's synthesis, formalized through CLI 3.0, proves empirically superior to either extreme position.

CLI 3.0 is not “another index” in political science. It is the mathematical formalization of a philosophical thesis about causal levels in evolutionary systems. It connects philosophy of biology with comparative law, memetics with linear algebra, Gould with Dawkins. If the framework survives expanded empirical testing ($n = 60+$, international validation, dynamic specification), it will have demonstrated that subtle conceptual distinctions, MLS1 versus MLS2, constraint versus selection, carry massive practical consequences. They distinguish

countries that can reform themselves from countries condemned to institutional stagnation. For the 88% of humanity living in non-WEIRD contexts where formal institutions chronically fail to produce their intended effects, that distinction is not academic.

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Author's declaration: The theoretical frameworks and interpretations in this paper are my sole intellectual work. Computational assistance was used for bibliographic formatting and mathematical notation. The Showan Khurshid framework is cited with the author's permission from his published work; the proposed extension of MRSP/MSSP to constitutional lock-in thresholds is my own formulation and does not represent Khurshid's claims. Any errors are my own.